

● MEDIUM

● ADVANCED MATH

Equivalent expressions

7 questions · ~11 min

04

Q1

ADVANCED MATH

Which of the following is equivalent to the sum of $3x^4 + 2x^3$ and $4x^4 + 7x^3$?

- A. $16x^{14}$
- B. $7x^8 + 9x^6$
- C. $12x^4 + 14x^3$
- D. $7x^4 + 9x^3$

Q2

ADVANCED MATH

$$(x - 11y)(2x - 3y) - 12y(-2x + 3y)$$

Which of the following is equivalent to the expression above?

- A. $x - 23y$
- B. $2x^2 - xy - 3y^2$
- C. $2x^2 + 24xy + 36y^2$
- D. $2x^2 - 49xy + 69y^2$

Q3

ADVANCED MATH

Which of the following is an equivalent form of $(1.5x - 2.4)^2 - (5.2x^2 - 6.4)$?

- A. $-2.2x^2 + 1.6$
- B. $-2.2x^2 + 11.2$
- C. $-2.95x^2 - 7.2x + 12.16$
- D. $-2.95x^2 - 7.2x + 0.64$

Q4

ADVANCED MATH

Which expression represents the product of $x^{-6}y^3z^5$ and $x^4z^5 + y^8z^{-7}$?

- A. $x^{-2}z^{10} + y^{11}z^{-2}$
- B. $x^{-2}z^{10} + x^{-6}z^{-2}$
- C. $x^{-2}y^3z^{10} + y^8z^{-7}$
- D. $x^{-2}y^3z^{10} + x^{-6}y^{11}z^{-2}$

$$4a^2 + 20ab + 25b^2$$

Which of the following is a factor of the polynomial above?

- A. $a + b$
- B. $2a + 5b$
- C. $4a + 5b$
- D. $4a + 25b$

Q6

GRID-IN ADVANCED MATH

The sum of $-2x^2 + x + 31$ and $3x^2 + 7x - 8$ can be written in the form $ax^2 + bx + c$, where a , b , and c are constants. What is the value of $a + b + c$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$gx = \frac{3}{5}x + \frac{7}{6}$$

$$hx = 6x - 5$$

The functions g and h are defined by the equations shown. Which expression is equivalent to $gx \cdot hx$?

A. $\frac{18x^2}{5} - \frac{35}{6}$

B. $\frac{18x^2}{5} + \frac{27x}{11} - \frac{35}{6}$

C. $\frac{18x^2}{5} - 4x - \frac{35}{6}$

D. $\frac{18x^2}{5} + 4x - \frac{35}{6}$